

What Happens Next? Closed- and Open-World Video Action Recognition

on a 33-Class Something-Something V2 Subset

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Abstract

We present a two-track study of temporal action prediction on a 33-class subset of Something-Something V2 (SSv2) [Goyal et al.(2017)Goyal, Ebrahimi Kahou, Michalski, Materzynska, Westphal, Kim, Haenel, Freund, Yianilos, Mueller, and Schiele]. “What Happens Next?”: only the first $\approx 40\%$ of each clip (four JPEG frames) is given, so the outcome is withheld and classification must reason about motion. In **Track A (Closed World)**, where pretraining is forbidden, two data-centric fixes dominate architecture choice: a 10-epoch *regularisation warm-up* (+36.3 pp) and matching the model’s frame count to the true clip length (+4.4 pp). Once these are fixed, a controlled ablation study shows that *where* a model mixes time matters more than *how much* capacity it has: Temporal-Shift-Module [Lin et al.(2019)Lin, Gan, and Han] (TSM), which mixes frames *before* spatial pooling, beats VideoFormer-Lite (VFL, global post-pooling attention) by +3.0 pp at matched backbone size, and scaling its backbone from ResNet18 to ResNet50 then adds a further +3.4 pp (**42.85% val_dir**, our best single model). A five-model log-softmax ensemble of TSM and VFL variants with test-time augmentation reaches **46.75% val_dir** and **0.5072** public / **0.4764** private

on Kaggle — our best closed-track submission.

In **Track B (Open World)**, we fine-tune V-JEPA 2 [Bardes et al.(2025)Bardes, Garrido, Ponce, Rabbat, LeCun, and VideoMAE [Tong et al.(2022)Tong, Song, Wang, and Wang]

backbones with progressive unfreezing, layer-wise learning-rate decay, window-capped source-frame re-extraction and pseudo-labelling, then build a 14-model uniform ensemble reaching **0.6586** Kaggle top-1 (16th in both tracks). We document

a self-discovered data-leakage event (a discarded 0.81 submission) and show that ensemble diversity across architecture families beats single-family or learned-weight ensembling. Failure analysis attributes the residual errors to real-vs-pretended and direction-ambiguous classes that four frames under-determine.

1 Problem Description

Task. The challenge classifies four-frame JPEG clips into one of 33 motion-defined action classes (e.g. *Pouring something into something*, *Pulling something from left to right*). Crucially, each clip contains only the *anticipatory* portion of the source SSv2 video—the shipped **frame_003** sits at a median fraction of 0.40 of the source clip—so the action’s outcome is deliberately withheld and the task probes motion prediction, not final-state recognition [Goyal et al.(2017)Goyal, Ebrahimi Kahou, Michalski, Materzynska, Westphal, Kim, Haenel, Freund, Yianilos, Mueller, and Schiele].

Dataset. The split provides 44,993 training clips, 6,745 validation clips and 6,913 test clips over 33 classes (numeric prefixes 000–032; class 027 is absent from training). Classes are heavily imbalanced: the largest (009, *Moving something up*) has 3,170 examples while the smallest (026, *Spilling something next to something*) has only 162, a ratio of 19.6×. The KL divergence between training and validation class distributions is 0.088, confirming the splits are representative.

Evaluation and tracks. Submissions are scored by top-1 accuracy on a public+private Kaggle leaderboard over two stages (Stage 1, intermediate; Stage 2, final); top-5 is tracked internally. The provided baseline reaches ≈ 0.55 private and the second-place public reference is 0.74; our final submissions placed **16th in both tracks** (counting the multiple accounts fielded by some teams). Following the challenge, we report two regimes. **Track A (Closed World)** prohibits pretrained models and external data—all spatial features, temporal dynamics and class boundaries are learned from scratch on the four provided frames. **Track B (Open World)** permits pretrained backbones and external corpora but not future frames or test labels. The task is hard for three reasons: (i) labels are motion-defined, defeating appearance-only models [Goyal et al.(2017)Goyal, Ebrahimi Kahou, Michalski, Materzynska, Westphal, Kim, Haelen, Freund, Yianiilos, Mueller-F]; (ii) four frames is an order of magnitude below the 8–32 standard in video recognition [Lin et al.(2019)Lin, Gan, and Han, Bertasius et al.(2021)Bertasius, Wang, and Torresani]; (iii) the discriminative cue—the action’s outcome—is withheld by construction.

2 Methods

2.1 Track A — Closed World

The pipeline evolved through several iterations on an internal training-split, all trained from scratch (the closed track forbids pretraining and external data). Our guiding hypothesis is that, without pretraining, *every parameter spent on temporal modelling*

is one not spent learning spatial features, favouring parameter-free temporal mechanisms over heavy temporal heads — and that *where* in the network temporal information is mixed matters as much as how it is mixed.

2.1.1 Architecture family and complexity

All three Track A architectures share the input/output contract $(B, T, C, H, W) \rightarrow (B, 33)$ with $T=4$, $H=W=224$, and build on a ResNet backbone [He et al.(2016)He, Zhang, Ren, and Sun] that maps each $224 \times 224 \times 3$ frame to a 512-d (ResNet18) or 2048-d (ResNet50) feature after global average pooling. They differ only in *how* the T per-frame feature vectors are turned into a single 33-way prediction — the choice that defines each model’s parameter budget and its temporal inductive bias.

CNNBaseline and **CNN-LSTM** simply average (resp. recurrently combine) the T pooled features; with four frames the LSTM gives only marginal gains because backpropagation through time on such short sequences is unstable. Both serve as lower bounds (≈ 11.2 M parameters, ResNet18 backbone).

TSM-ResNet [Lin et al.(2019)Lin, Gan, and Han] wraps every residual block: a fraction $1/\text{fold_div}$ of channels is shifted one step backward and another $1/\text{fold_div}$ one step forward along the time axis,

$$\text{TSM}(\mathbf{x})_{t,c} = \begin{cases} \mathbf{x}_{t-1,c} & c < C/4 \\ \mathbf{x}_{t+1,c} & C/4 \leq c < C/2 \\ \mathbf{x}_{t,c} & \text{otherwise,} \end{cases} \quad (1)$$

mixing temporal information at *zero parameter cost* by re-indexing tensor slices *before* every convolution, while spatial resolution is still full. After 8 such blocks, a 512-d (R18) or 2048-d (R50) per-frame vector is produced, the T vectors are averaged, and a linear head $d \times 33$ yields the logits. Total parameter count is therefore *exactly* that of the backbone plus the head: ≈ 11.2 M for ResNet18 ($P_{\text{head}}=512 \times 33 \approx 17$ K) and ≈ 23.6 M for ResNet50 ($P_{\text{head}}=2048 \times 33 \approx 68$ K, of which 23.5M sit in the backbone). TSM was designed for SSv2-style motion labels and is therefore a strong prior here (Fig. 1).

Technique	Setting	Source	The ensemble reaches 46.75% on the official <code>val_dir</code> (Fig. 6) and 0.5072 public / 0.4764 private on Kaggle — our best closed-track leaderboard score.
Reg. warm-up	10 ep, no MixUp/CutMix	this work	+9.1 pp public over the strongest single-model submission (tsm.ultra.v2 + TTA, 0.4163; Table 2). 10-crop TTA alone contributes ~1 pp on the strongest single-model (Fig. 8) with Li and Shih [2019].
MixUp / CutMix	$\alpha=0.2$ / $p=0.25$, after ep 10	[Zhang et al.(2018)Zhang, Cisse, Dauphin, and Lopez-Paz, Yun et al.(2019)Yun, Han, Kim, and Yoo]	
Focal loss	$\gamma \in \{0, 1, 2\}$ (best ≈ 1)	[Lin et al.(2017)Lin, Goyal, Girshick, He, and Dollár]	
Label smoothing	$\varepsilon=0.1$	[Müller et al.(2019)Müller, Kornblith, and Hinton]	
RandAugment	2 ops, mag. 0.5	[Cubuk et al.(2020)Cubuk, Zoph, Shlens, and Hinton]	
Random erasing	$p=0.25$	[Zhong et al.(2020)Zhong, Wang, Li, and Shih]	
Horizontal flip	disabled		direction-encoded classes (Fig. 9) where the 5 hflip views push some confidences toward the minor class even after label-aware logit remapping.
Stochastic depth	DropPath 0.1	[Huang et al.(2016)Huang, Sun, Lid, Sedra, and Weinberger]	
Optimiser	AdamW, lr 10^{-3} , cosine	[Loshchilov and Hutter(2019)]	

Table 1: Track A training recipe (best single model: TSM-ResNet50, $T=4$, `fold.div=8`, rotating folds, $\sim 42.85\%$ `val_dir`).

left→right); CutMix probability is kept low ($p=0.25$) because a cut patch spans all four frames and frequently occludes the hand/object motion that defines the label.

2.1.4 Rotating k -fold and ensemble

Beyond the standard train/val split, the strongest Track A checkpoints have a *rotating-fold* variant that is retrained on every fold of the train+val union (held-out fold = epoch mod k , $k=5$ by default), providing higher-bias snapshots that nonetheless contribute uncorrelated errors at inference (Fig. 10). This protocol adds +1.2 pp for TSM-ResNet18 and +4.3 pp for VFL — VFL benefits more because its global-attention head is data-hungry and lacks the structural temporal prior that TSM’s shift provides; an ablation on the fold count (Sec. 3.1) shows VFL is already saturated at $k=5$.

The final closed-track submission fuses *five* checkpoints by log-softmax late fusion. Three use 10-crop spatial TTA (5 crops $\times$ 2 flips): `tsm.ultra.v2` (57.73% internal), its rotating-fold variant, and `video_former_lite_ultra_rot`. Two run without TTA (inference cost): the new **TSM-ResNet50 rotating** (42.85%, our best single model) and a CE-trained ($\gamma=0$) TSM ablation that adds complementary calibration. Averaging in log-probability space down-weights over-confident errors and rewards agreement, suiting members of different calibration.

2.2 Track B — Open World

Top SSv2 teams use video transformers pre-trained on large unlabelled corpora. We selected two self-supervised families: **Video-MAE** [Tong et al.(2022)Tong, Song, Wang, and Wang] (masked-autoencoder pre-training, building on image MAE [He et al.(2022)He, Chen, Xie, Li, Dollár, and Girshick]/BEiT [HuggingFace Kinetics-400 checkpoints]) and **V-JEPA 2** [Bardes et al.(2025)Bardes, Garrido, Ponce, Rabbat, LeCun, a joint-embedding predictive architecture that reconstructs masked regions in *feature* space rather than pixel space, focusing capacity on semantically relevant motion. We use the self-supervised `vjepa2-vit1-fpc64-256` checkpoint, pretrained on unlabelled video without any SSv2 labels—a requirement for integrity (Sec. 3.4).

Progressive unfreezing with LLRD.

We fine-tune V-JEPA in three phases with layer-wise learning-rate decay (LLRD) [Howard and Ruder(2018), Clark et al.(2020)Clark, Luong, Le, and Manning, Bao et al.(2022)Bao, Dong, Piao, and Wei], where lower encoder block ℓ receives $\eta \cdot \lambda^\ell$ with $\lambda=0.75$. *Phase 0*: a frozen attentive probe (head only) reaches 0.672 EMA internal val. *Phase 1*: unfreezing the top-6 blocks with LLRD reaches 0.705. *Phase 2*: full-encoder fine-tuning reaches 0.728 internal and 0.6411 `val_dir`. Freezing first stabilises the head and preserves the pretrained feature geometry before large gradients reach lower layers; the characteristic staircase — each phase warm-starting above the previous plateau — is visible in Fig. 12. (These

training phases are distinct from the challenge’s Stage 1/Stage 2 leaderboard milestones.)

Window-capped source-frame re-extraction.

V-JEPA benefits from 16 input frames, but only four are shipped. Because folder names encode SSv2 video IDs, the full source WebM files are accessible. We re-extract 16 frames per clip in a *window-capped* manner: the shipped `frame_003` is matched against source frames by pixel-level mean absolute error to locate the end of the provided window, 16 frames are sampled uniformly within $[0, \text{window_end}]$, and the search is hard-bounded to the first 50% of the source clip so the withheld outcome ($> 60\%$) is unreachable by construction (median window fraction 0.39, max 0.50; Fig. 13).

Pseudo-labelling. Following noisy-student self-training [Xie et al.(2020)Xie, Luong, Hovy, and Lee, Lee(2013)], the Phase-2 snapshot ensemble generated soft labels for the test clips; predictions with max probability ≥ 0.85 were added as pseudo-labels (the model’s over-confident calibration meant that this threshold retained all 6,913 test clips, so pseudo-labelling effectively augments the train set with the entire test distribution under V-JEPA-ft’s own predictions). The model is then continue-finetuned on $\text{train} \cup \text{pseudo-test}$. The resulting V-JEPA-pseudo reaches 0.6519 `val_dir` / 0.6410 Kaggle, but the widened val/Kaggle gap (Sec. 3) signalled over-fitting to the validation distribution, so pseudo-labelling was not iterated.

Honest cross-preprocessing ensemble. The final Track B system aggregates 14 checkpoints from three families, each fine-tuned with its own preprocessing: V-JEPA (6 snapshots, 16-frame 256 px window-capped), VideoMAE-Base K400 (3, 4-frame 224 px), VideoMAE-Large K400 (3, 4-frame 224 px) and TSM-ResNet50 (2, 4-frame). Inference runs each model on its own preprocessing chain; per-model softmax tensors are then combined by *uniform* averaging. The key finding is that uniform averaging of diverse honest models outperforms learned or manual weighting (Sec. 3): architecturally distinct models produce

uncorrelated errors that average out, so even weaker members raise the ensemble.

2.3 Reused vs. own code

We *reuse*: the course starter repository (the Hydra configuration system, the `VideoFrameDataset` loader and the `train.py` skeleton); the torchvision ResNet [He et al.(2016)He, Zhang, Ren, and Sun] and R(2+1)D [Tran et al.(2018)Tran, Wang, Torresani, Ray, LeCun, and Fergus] backbones; and the pretrained HuggingFace V-JEPA 2 [Bardes et al.(2025)Bardes, Garrido, Ponce, Rabbat, LeCun, and Wang] and VideoMAE [Tong et al.(2022)Tong, Song, Wang, and Wang] checkpoints (Track B only). Our *own* implementations are: the parameter-free TSM block wrapper (re-implemented from the description of Lin et al. [Lin et al.(2019)Lin, Gan, and Han]); VideoFormer-Lite, our CNN+Transformer temporal head inspired by ViT [Dosovitskiy et al.(2021)Dosovitskiy, Beyer, Kolesnikov, Weissenborn, and Zhai] and pre-LN Transformers [Xiong et al.(2020)Xiong, Yang, He, Zheng, Zheng, Xing, Zhang, and He]; the regularisation-warm-up recipe, the rotating k -fold protocol, the controlled ablation suite (focal- γ , VFL depth/fold-count/schedule sweeps), and the log-softmax ensemble / 10-crop TTA pipeline; and the full Track B stack—window-capped source-frame re-extraction, the progressive-unfreezing + LLRD fine-tuning wrapper (following the layer-wise decay practice of [Howard and Ruder(2018), Clark et al.(2020)Clark, Luong, Le, and Manning, Bao et al.(2022)Bao, Dong, Piao, and Wei]), and the noisy-student pseudo-labelling loop [Xie et al.(2020)Xie, Luong, Hovy, and Lee, Lee(2013)].

3 Results

Overall. Table 2 reports headline results, now with Kaggle scores for *both* tracks. Track A reaches **46.75%** `val_dir` / **0.5072** Kaggle public from scratch; Track B reaches **0.6586** Kaggle top-1 (internal val 65.19%) with the 14-model uniform ensemble—well above the 0.55 provided baseline and

Trk	Method	val dir	Kaggle pub	temp. made
A	TSM-ResNet18 (single, +TTA)	41.68%	0.4163	0.3881
A	TSM-ResNet18 (single)	38.25%	0.3576	0.3442
A	VFL-Ultra (single)	32.11%	—	—
A	TSM-ResNet50 rot. (single)	42.85%	—	—
A	3-model ens. + TTA (closed)	44.03%	0.4508	0.4260
A	5-model ens. (R50)	46.75%	0.5072	0.4764
B	V-JEPA-ft standalone	64.11%	0.6361	0.6110
B	V-JEPA-pseudo	65.19%	0.6410	0.6110
B	14-uniform	65.19%	0.6586	0.6586

Table 2: Summary of Track A and Track B results (val = internal/val.dir top-1; Kaggle = public / private leaderboard top-1, reported separately for Track A where the gap is informative, and as a single combined figure for Track B as in the original submissions). The Track A ResNet50 single model has no individual Kaggle submission — it is evaluated only as an ensemble member.

approaching the 0.74 second-place public reference. The ≈ 15 pp Kaggle gap between the two tracks’ best submissions quantifies the practical value of large-scale self-supervised pretraining, unavailable in the closed track; the gap is markedly smaller than the ~ 18 -pp val.dir gap because Track A’s ensemble generalises comparatively better from val.dir to the Kaggle test distribution (see also the calibration discussion below).

3.1 Track A ablations

Regularisation warm-up (+36.3 pp). Identical architecture and config, single variable: without warm-up VFL stalls at 9.45% (killed at epoch 4) and R(2+1)D [Tran et al.(2018)Tran, Wang, Torresani, Ray, LeCun, and Poluri] at 10.78%; with a 10-epoch warm-up VFL reaches 45.78%. The effect transfers across architectures, indicating a general property of cold-start training with label-mixing rather than an architecture-specific quirk.

Frame count (+4.4 pp). $T=5 \rightarrow 4$ improves internal val by +4.41 pp (53.32 $\rightarrow$ 57.73) and val.dir by +3.34 pp (34.91 $\rightarrow$ 38.25); the *first* ResNet50 at-

tempt made at the old $T=8$, regressed 8 pp below the $T=4$ ResNet18 internal score (Fig. 3).

Architecture: TSM vs. VFL (+3.0 pp for local mixing). At matched settings (rotating folds, $T=4$, ResNet18 backbone, no TTA), TSM reaches 39.45% val.dir vs. VFL’s 36.44% (+3.01 pp). This is the controlled test of the architectural argument in Sec. 2.1.1: with backbone, frame count, recipe and protocol held fixed, the only difference is *where* the two models mix time — before pooling (TSM) or after (VFL) — and the local-mixing model wins by a clear margin, consistent with SSv2 labels being defined by small, spatially-localised motion cues that pooling destroys.

Backbone scale-up, done right (+3.4 pp).

Once $T=4$ and rotating folds are fixed, replacing the ResNet18 backbone with ResNet50 (11.2 $\rightarrow$ 23.6 M parameters, Sec. 2.1.1) brings 39.45% $\rightarrow$ 42.85% val.dir (+3.40 pp, Fig. 2). Reading this together with the $T=8$ regression above closes the loop on our central architectural claim: *capacity is useless on top of a corrupted temporal signal, but valuable once the signal is correct* — the $2.1\times$ parameter increase here converts cleanly into accuracy, whereas the earlier $T=8$ run wasted the same increase on duplicated frames.

Focal-loss γ sweep (+0.5–0.6 pp, optimum flat).

With every other variable frozen (rotating folds, TSM-ResNet18), CE ($\gamma=0$) reaches 38.92%, $\gamma=1$ reaches 39.50%, and $\gamma=2$ reaches 39.45% (Fig. 4). Any focal weighting beats CE by ≈ 0.5 pp; the $\gamma=1$ vs. $\gamma=2$ gap (0.05 pp) is within run-to-run noise, so the optimal γ lies in $[1, 2]$ and its exact value is not critical — evidence that the gain comes from *having* a confusable-pair-aware loss, not from tuning it finely. Despite scoring lowest standalone, the CE model contributes +0.52 pp to the 5-model ensemble (Sec. 2.1.4): its different loss geometry yields a different confidence calibration, which is exactly the kind of complementary error the log-softmax fusion exploits.

VFL depth and fold count. Reducing VFL from 3 to 2 Transformer layers (Eq. (2): $P_{\text{Transformer}}$ drops from 9.4 $\rightarrow$ 6.3 M) costs 36.44% $\rightarrow$ 35.36% (−1.08 pp): the third block provides a real, if modest, gain at this data scale (Fig. 5). Increasing the rotation from 5 to 10 folds (90% vs. 80% of the data seen per epoch) instead costs −0.46 pp (36.44 $\rightarrow$ 35.98%): VFL has already saturated its data-utilisation capacity at $k=5$, so additional coverage trades off against noisier gradient estimates from smaller per-fold validation sets. Switching the SGDR restart shape ($T_{\text{mult}} 1 \rightarrow 2$) changes accuracy by at most −0.12 pp for either architecture — the rotating-fold protocol already supplies enough implicit regularisation that the exact LR-restart geometry is largely irrelevant.

Ensemble components and leave-one-out. Table 3 (Fig. 7) shows the leave-one-out analysis on the final 5-model ensemble (46.75%). Every member is necessary, but the *margins* are revealing: removing the ResNet50 model costs −1.33 pp — by far the largest single drop, larger than removing any same-backbone TSM-ResNet18 variant (−0.31 to −0.47 pp) or even the architecturally-different VFL (−0.64 pp). **Backbone diversity contributes more to the ensemble than architecture-family diversity at matched capacity.** This refines the earlier (3-model) finding that TSM and VFL are complementary: that remains true, but the single largest source of decorrelated, useful error in our pool is a model that simply *sees more* (deeper backbone), not one that *sees differently*.

3.2 Track B ablations

Honest Kaggle progression. Table 4 (with Fig. 11 as its visual counterpart) traces every clean submission. The 21 pp gap between the discarded leaky run (0.81) and the first honest baseline (0.6361) quantifies what data leakage was buying; every subsequent gain was earned by legitimate architectural diversity (Fig. 3 is the Track A counterpart). Fig. 14 plots Kaggle vs. ensemble size: returns are still positive at $n=14$, and the $n=6$ V-JEPA-only configuration underperforms the $n=9$ uniform mix despite higher individual accuracies.

Configuration	Top-1	Δ
Full ensemble (5 models)	46.75%	—
– TSM-ResNet50 (rot)	45.41%	−1.33
– VFL-Ultra-rot +TTA	46.11%	−0.64
– TSM-no-focal (rot)	46.23%	−0.52
– TSM-Ultra-v2-rot +TTA	46.27%	−0.47
– TSM-Ultra-v2 +TTA	46.43%	−0.31

Table 3: Track A leave-one-out on `val_dir` for the final 5-model ensemble (46.75%). Every component is necessary; the ResNet50 backbone is by far the dominant contributor, ahead of the architecturally distinct VFL.

Submission	Kaggle	Δ
Leaky V-JEPA (discarded)	0.81	—
V-JEPA-ft standalone (clean)	0.6361	—
Ensemble v1 (9-uniform)	0.6418	+0.6
V-JEPA-pseudo standalone	0.6410	+0.5
V-JEPA-only (6 snapshots)	0.6410	regress.
V-JEPA-heavy (3:1 manual)	0.6499	+1.4
12-uniform (+ pseudo)	0.6546	+1.9
14-uniform (+ Large)	0.6586	+2.3

Table 4: Track B Kaggle submission history (Δ in pp vs. the first honest baseline). Diversity, not single-family scaling, drives every gain.

Diversity beats weighting. Table 5 isolates ensemble strategy on a fixed checkpoint pool. A 6-snapshot V-JEPA-only ensemble (0.6410) *underperforms* a 12-model uniform mix (0.6546) despite V-JEPA’s higher individual accuracy; manual 3:1 weighting toward V-JEPA (0.6499) falls between the two. Uniform averaging of architecturally distinct families wins because their errors are uncorrelated. Table 6 makes this concrete: TSM-ResNet50, though weakest standalone (0.3364 `val_dir`, Table 7), has the lowest pairwise agreement with every transformer family (0.37–0.38), whereas two V-JEPA variants agree 0.85—so a V-JEPA-only ensemble is near-redundant. Fig. 15 shows the failure mode of learned weighting on the same pool: a softmax-weighted fit collapses $\sim 82\%$ of mass onto the two strongest checkpoints and effectively zeroes K400, throwing away

Strategy	n	Kaggle
V-JEPA-only, uniform	6	0.6410
9-uniform (+ K400 + TSM)	9	0.6418
V-JEPA-heavy (3:1 weight)	12	0.6499
12-uniform	12	0.6546
14-uniform	14	0.6586

Table 5: Track B ensemble-strategy ablation. Uniform averaging of diverse families beats single-family and learned/manual weighting.

Family A	Family B	Agreement
V-JEPA-pseudo	TSM	0.370
V-JEPA-ft	TSM	0.372
K400-Base	TSM	0.375
V-JEPA-ft	K400-Base	0.582
V-JEPA-ft	V-JEPA-pseudo	0.854

Table 6: Pairwise inter-family prediction agreement on the test set. Lower agreement implies more complementary (uncorrelated) errors.

the very decorrelation that the uniform ensemble exploits. This mirrors, at a larger scale, our Track A finding (Sec. 3.1) that the most valuable ensemble member is the one that is most *different*, not the one that is individually strongest.

Calibration. The `val_dir`→Kaggle gap is 0.005 for V-JEPA-ft but 0.011 for V-JEPA-pseudo—a doubling that warned us against further pseudo-label iterations. By contrast, Track A’s best ensemble has a *negative* gap (46.75% `val_dir` vs. 0.5072 Kaggle public, i.e. Kaggle is *higher*): a from-scratch model trained with heavy augmentation is, if anything, slightly better calibrated to an unseen test distribution than to its own (smaller, finite) validation set, whereas a fine-tuned pretrained model can specialise to `val_dir` quirks. The train-split→Kaggle gap for Track B (≈ 0.09) reflects that the internal split is drawn from the training distribution while the Kaggle test set is sampled independently.

Family	Snapshots	<code>val_dir</code> top-1
VideoMAE-Base K400 (4f)	3	0.5576
VideoMAE-Large K400 (4f)	2	0.5945
VideoMAE-Base SSv2 (4f)	3	0.6047
TSM-ResNet50 (4f)	3	0.3364
Uniform ensemble	11	0.6222

Table 7: Per-family single-model `val_dir` top-1 (Track B, single-view). No single family reaches the ensemble; the weak TSM family still adds value through uncorrelated errors.

3.3 Failure analysis

Both tracks share the same dominant error patterns, which mirror SSv2’s design intent (Figs. 16, 17, 18). *(i) Real vs. pretended.* The worst single confusion is *Picking something up* predicted as *Pretending to pick something up* (56/199 clips; F1 = 0.07 in Track A): the early-clip trajectory is identical and the disambiguating cue—whether the object actually rises—falls in the withheld future. *(ii) Temporal direction.* *Folding*↔*Unfolding* (24–29 mutual errors) share spatial appearance and differ only in motion direction, fragile at four frames. *(iii) Static sink.* *Holding something* absorbs misclassified motion clips (*Moving down*: 30; *Opening*: 28), acting as the default when no decisive motion is detected. The best classes (F1 > 0.54) are all clean single-arc motions (*Pulling left→right*, *Pouring*, *Uncovering*, *Folding*). The correlation between class size and per-class accuracy is weak ($r=0.24$), so difficulty stems from semantic ambiguity, not scarcity. The top-5/top-1 gap on `val_dir` is large (0.882 vs. 0.622) and widest for the hardest classes (*Picking up*: +61.3 pp top-5), indicating models rank the correct class highly but place a visually similar “pretend” variant first.

3.4 Data-integrity

An early extraction run sampled frames uniformly over the full source clip (0–100%); training V-JEPA on these yielded 0.81 Kaggle, ~ 7 pp above second place (the leaky red bar in Fig. 11). An audit revealed two concurrent leakage vectors: (1) frames

from the withheld outcome ($> 60\%$) were reachable, giving the model the explicitly withheld information; (2) an initial V-JEPA checkpoint had been fine-tuned on SSv2 *with ground-truth labels*, so the backbone had seen the test videos. Both runs were discarded and the pipeline rebuilt with the self-supervised `vjepa2-vit1-fpc64-256` backbone and the 50% hard bound (Sec. 2.2); we treat the 21 pp honest/leaky difference as a direct measurement of the combined leakage. The 50% cap is below the $\approx 60\%$ ceiling that the rules explicitly mark as “available”, an additional safety margin against matching-noise overshoot rather than a rules requirement. Three infrastructure faults were also fixed: a silent NFS quota producing 0-byte checkpoints (resolved with `fsync-durable` atomic saves), a top- K snapshot rename race, and a cross-architecture ensembling bug that silently loaded Large weights into a Base graph (both `model_name`-tagged), fixed by rebuilding each model from its stored config at inference.

4 Conclusion and Future Work

We presented a two-track solution to video action anticipation on SSv2. **Track A** shows that, from scratch, *data-centric* fixes (a +36.3 pp regularisation warm-up, a +4.4 pp frame-count correction) dominate raw architecture choice — but once they are in place, a controlled ablation reveals a clear architectural ranking: local pre-pooling temporal mixing (TSM) beats global post-pooling attention (VFL) by +3.0 pp at matched capacity, and a $2.1\times$ deeper backbone then adds a further +3.4 pp (**42.85% val_dir**, our best single model). The resulting five-model log-softmax ensemble reaches **46.75% val_dir** and **0.5072** on the public Kaggle leaderboard, with leave-one-out analysis showing that backbone diversity (R50 vs. R18) contributes more to the ensemble than architecture-family diversity at equal capacity. **Track B** shows that progressive unfreezing with LLRD enables effective fine-tuning of large self-supervised video transformers (V-JEPA 2 [Bardes et al.(2025)Bardes, Garrido, Ponce, Rarhant, Guillaumeyan, Singh, and Balas], implemented the VideoMAE [Tong et al.(2022)Tong, Song, Wang, and Wang] Track B (Open-World) pipeline: the window-capped source-frame re-extraction, the V-JEPA 2

diversity across architecture families consistently outperforms single-family or learned-weight ensembling (0.6586 Kaggle) — the same principle, observed at a larger scale. A self-discovered leakage audit quantifies a 21 pp integrity effect and documents the honest rebuild. Failure analysis reveals a fundamental limit shared by both tracks: the dataset’s design intent—distinguishing real from pretended actions and forward from reverse motion—requires the withheld outcome, which four frames cannot supply.

Future work. (1) Running 10-crop TTA for TSM-ResNet50 (expected $\sim +2$ pp based on the ResNet18 measurement, currently impractical at submission time due to inference cost); (2) V-JEPA-Giant once inference-time memory is addressed; (3) targeted class rebalancing for the *holding/pretending* sinks in both tracks; (4) temperature-scaled confidence thresholds for a second, less over-fitting pseudo-label round; (5) a 15th Track B ensemble member from a 16-frame window-capped VideoMAE-Large checkpoint, completed after the leaderboard freeze, yielded 0.6592 (within noise of 0.6586, confirming returns have flattened); (6) explicit trajectory-prediction heads to attack the real-vs-pretended ambiguity directly; (7) applying the $T=4$ fix and rotating-folds protocol to the Track B TSM family, whose current weakness (0.3364 `val_dir`) likely stems from the same frame-count issue that cost Track A 4.4 pp.

Contributions

Andrea Signoretti designed and implemented the full Track A (Closed-World) pipeline: the TSM-ResNet and VideoFormer-Lite models, the Hydra training/ k -fold framework, the regularisation warm-up and frame-count analysis, the ResNet50 backbone scale-up, the controlled ablation suite (architecture, focal- γ , VFL depth/folds/schedule), and the log-softmax ensemble/TTA code, and produced the Track A ablation and per-class figures. **Florent Guillaumeyan** designed and implemented the Track B (Open-World) pipeline: the window-capped source-frame re-extraction, the V-JEPA 2

fine-tuning wrapper with progressive unfreezing and LLRD, the pseudo-labelling stage, the honest cross-preprocessing 14-model ensemble, and the data-integrity audit. Both authors jointly analysed results and wrote this report.

Figures

All figures are referenced from the main text (Secs. 2.1–3.3).

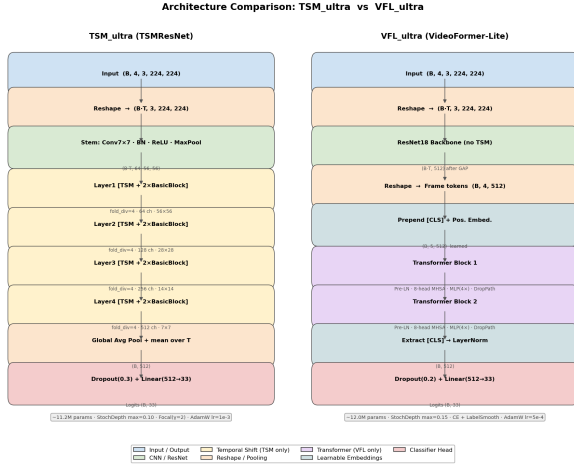


Figure 1: The two complementary Track A temporal mechanisms. **TSM** mixes time *locally* via parameter-free channel shifts in every ResNet block, *before* spatial pooling; **VideoFormer-Lite** mixes time *globally* via [CLS]-token attention over pooled frame descriptors. The controlled comparison in Sec. 3.1 shows the former wins by +3.0 pp at matched backbone size (Sec. 2.1).

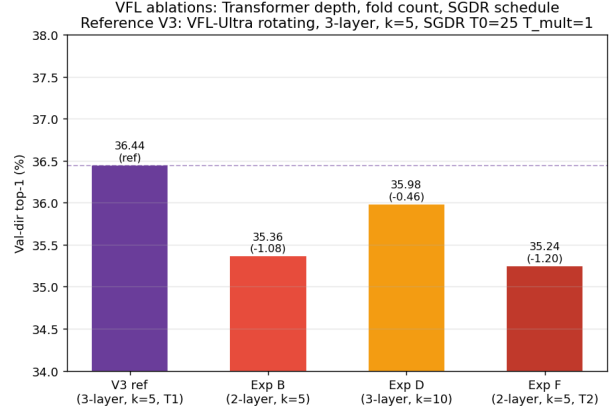


Figure 2: TSM backbone and frame-count progression on `val_dir`. The $T=8$ ResNet50 attempt regresses because interpolation duplicates frames; restoring $T=4$ and scaling to ResNet50 with rotating folds instead adds +3.4 pp over the matched ResNet18, reaching the new single-model best of 42.85% (Sec. 2.1).

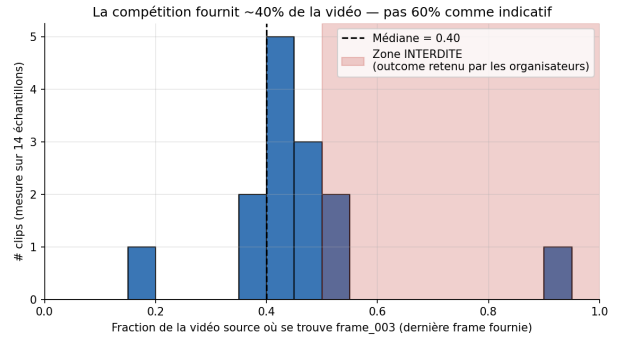
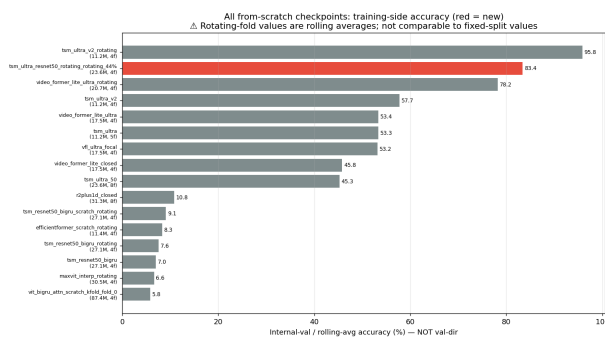
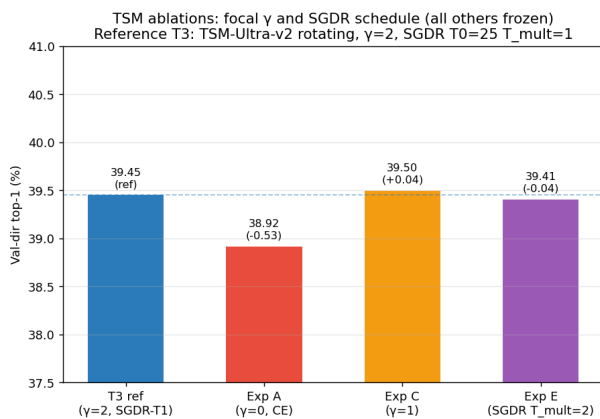
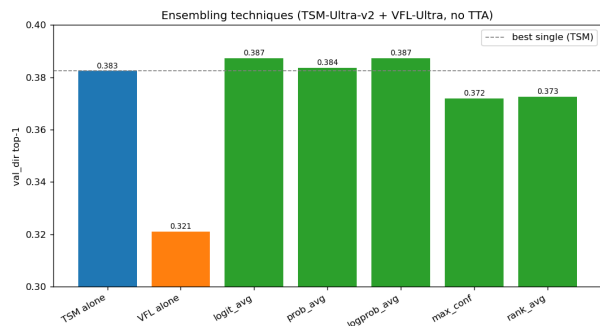


Figure 3: Per-model `val_dir` top-1/top-5 (Track A). Correcting the frame count (TSM-Ultra→TSM-Ultra-v2) and then scaling the backbone to ResNet50 (rotating folds) are the two largest per-model jumps; the $T=8$ ResNet50 scale-up regresses (Sec. 3).



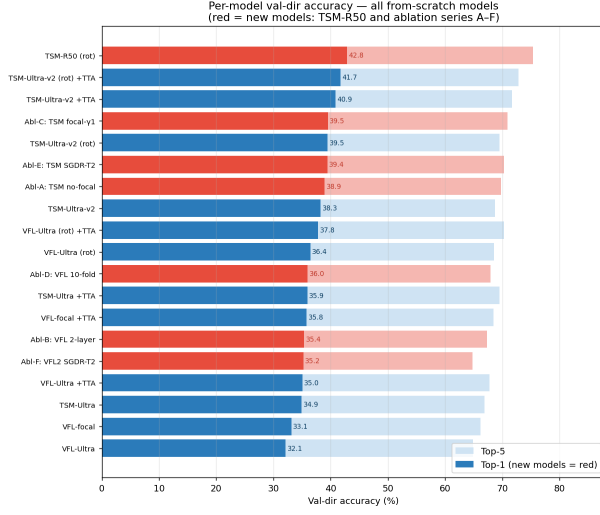


Figure 8: Effect of 10-crop spatial TTA on Track A models. TTA adds ~ 1 pp on the strongest checkpoint and is essential at the ensemble level, where independent random errors from different crops average out (Sec. 2.1).

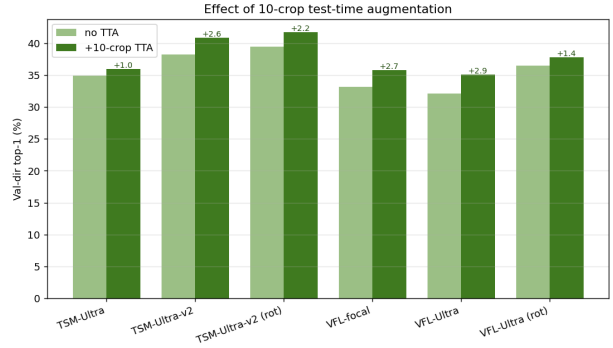


Figure 10: Rotating-fold training. Re-training the strongest checkpoints on every train/val fold yields snapshots that share weights but disagree on the boundary, contributing uncorrelated errors that the log-softmax ensemble exploits; the gain is larger for VFL (+4.3 pp) than for TSM (+1.2 pp) (Sec. 2.1.4).

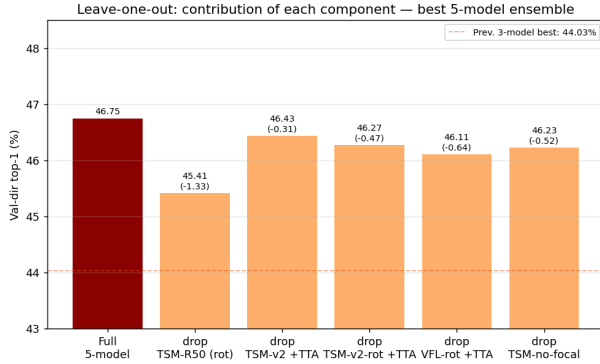


Figure 9: TTA directional-class trade-off. Horizontal-flip TTA improves most classes but hurts a few direction-encoded ones (*Pulling left $\rightarrow$ right* vs. *right $\rightarrow$ left*) even after the flip-aware logit remap, because mirror evidence is intrinsically ambiguous when motion is short.

VFL_ultra Architecture
(VideoFormer-Lite · ResNet18 + 2×Transformer)

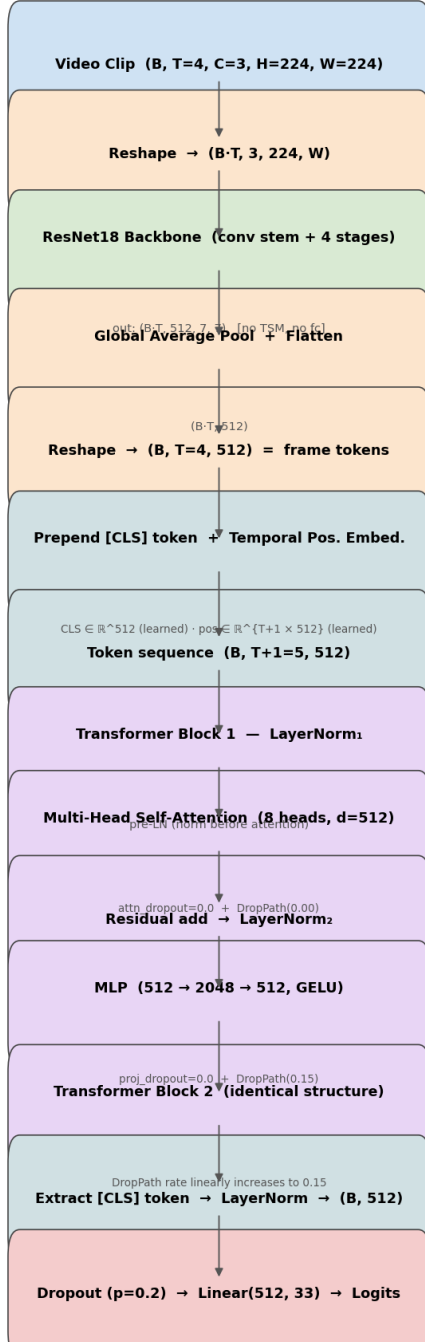


Figure 11: Track B honest Kaggle progression. The discarded leaky submission (0.81, red) is plotted alongside every clean ensemble; each gain comes from adding architecturally distinct members, not from re-weighting. The dashed line marks the second-place public reference (0.74); the gap to it bounds what self-supervised pretraining + diverse ensembling buys us (Sec. 3, Table 4).

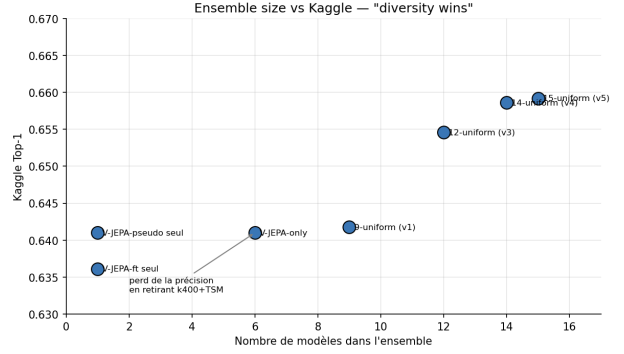


Figure 12: V-JEPA progressive unfreezing + LLRD (Track B internal val). Each phase warm-starts above the previous plateau: frozen probe (0.55 $\rightarrow$ 0.67), then top-6 unfrozen + LLRD (0.68 $\rightarrow$ 0.71), then full encoder (0.70 $\rightarrow$ 0.73), then pseudo-label retrain (0.72 $\rightarrow$ 0.74). The staircase pattern is the visual signature that LLRD has preserved each phase's feature geometry (Sec. 2.2).

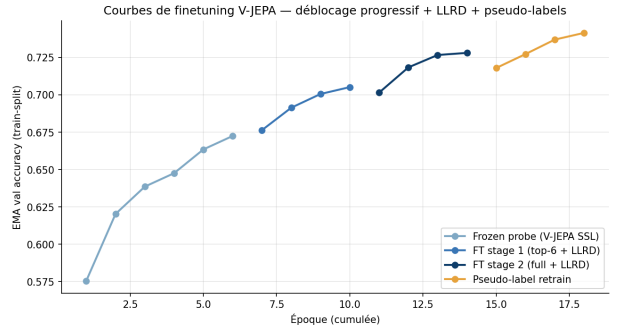


Figure 13: Where shipped `frame_003` sits in the source SSv2 clip (fraction). The dashed line at 0.40 is the median end-of-window; the shaded region [0.50, 1.00] is the withheld outcome — our re-extraction is hard-bounded to the un-shaded region (Sec. 3.4).

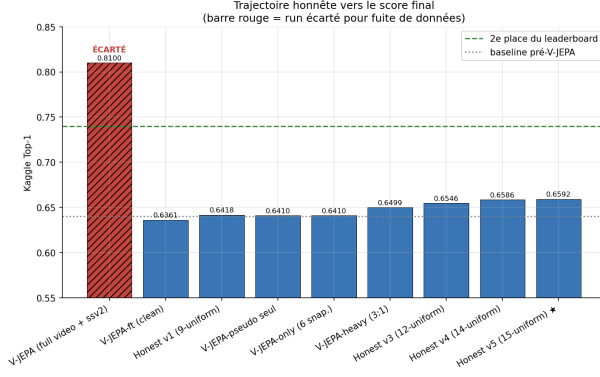


Figure 14: Track B Kaggle vs. ensemble size. Returns from adding diverse honest members are still positive at $n=14$; at $n=6$, dropping K400 + TSM *hurts* despite their lower individual scores, because their uncorrelated errors disappear from the average (Table 5).

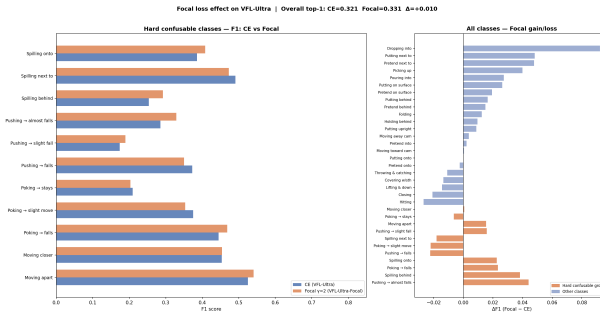


Figure 15: Learned ensemble weights collapse onto the two strongest members ($\sim 82\%$ of mass on `large_attempt1_top2` and `ssv2_top1`), effectively zeroing K400 contributions. This is exactly why a learned weighting fails to beat uniform: it discards the very decorrelation signal that diverse weaker members provide (Sec. 3).

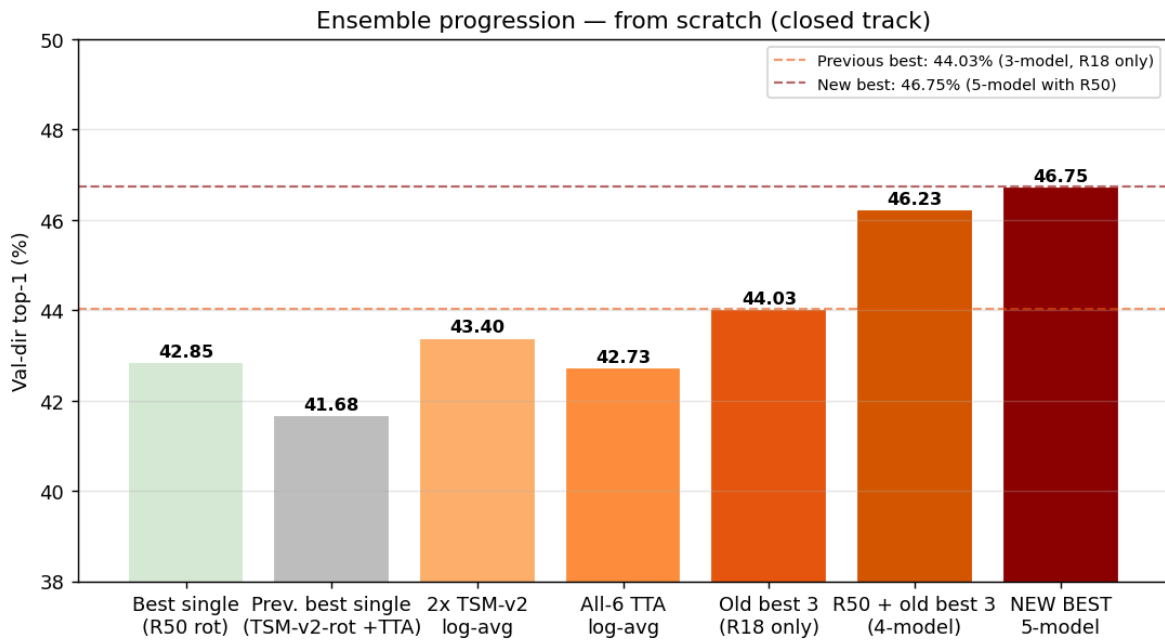


Figure 16: Per-class F1 (5-model ensemble, `val_dir`). Directional single-object motions score highest; “pretending” and direction-ambiguous classes score lowest (Sec. 3.3).

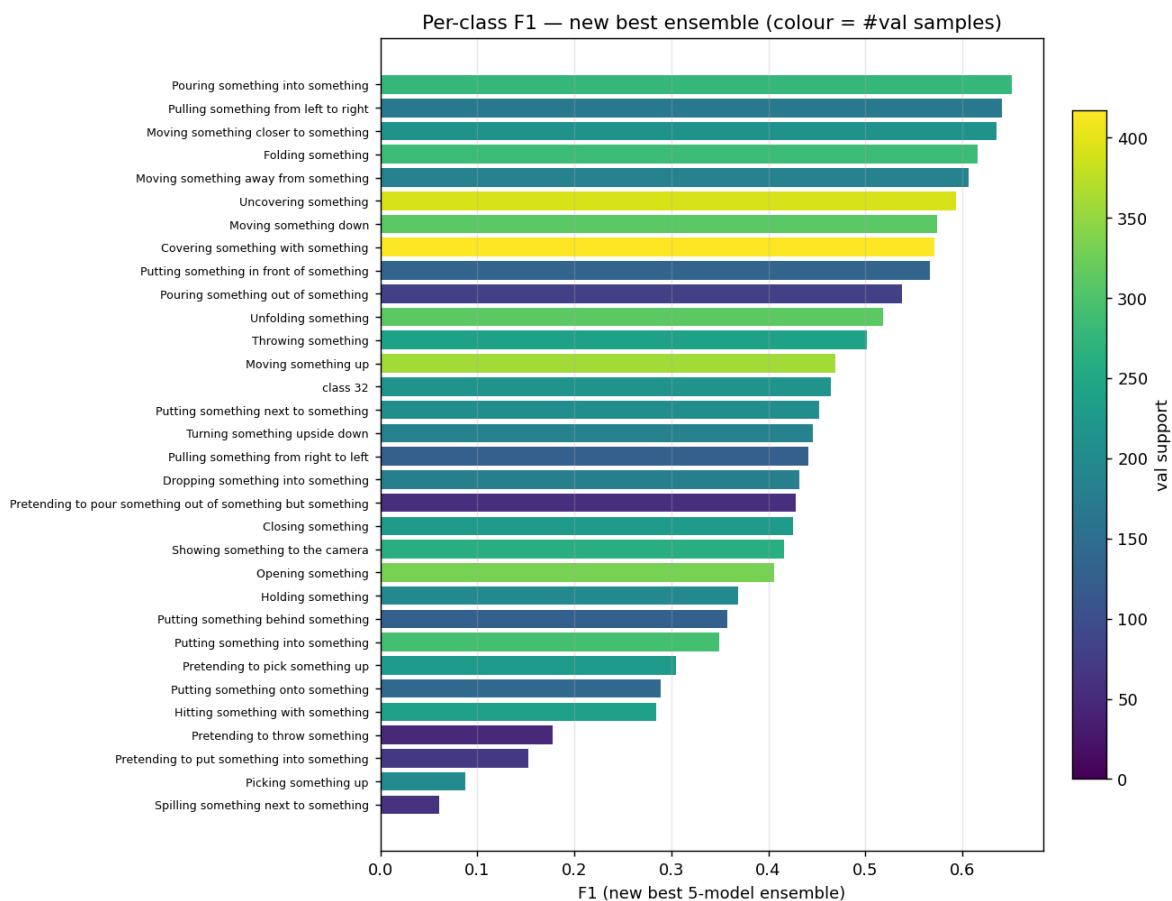


Figure 17: 33×33 confusion matrix (5-model ensemble, `val_dir`). Off-diagonal mass concentrates on semantically adjacent pairs (real vs. pretended; fold vs. unfold; motion vs. *holding*).

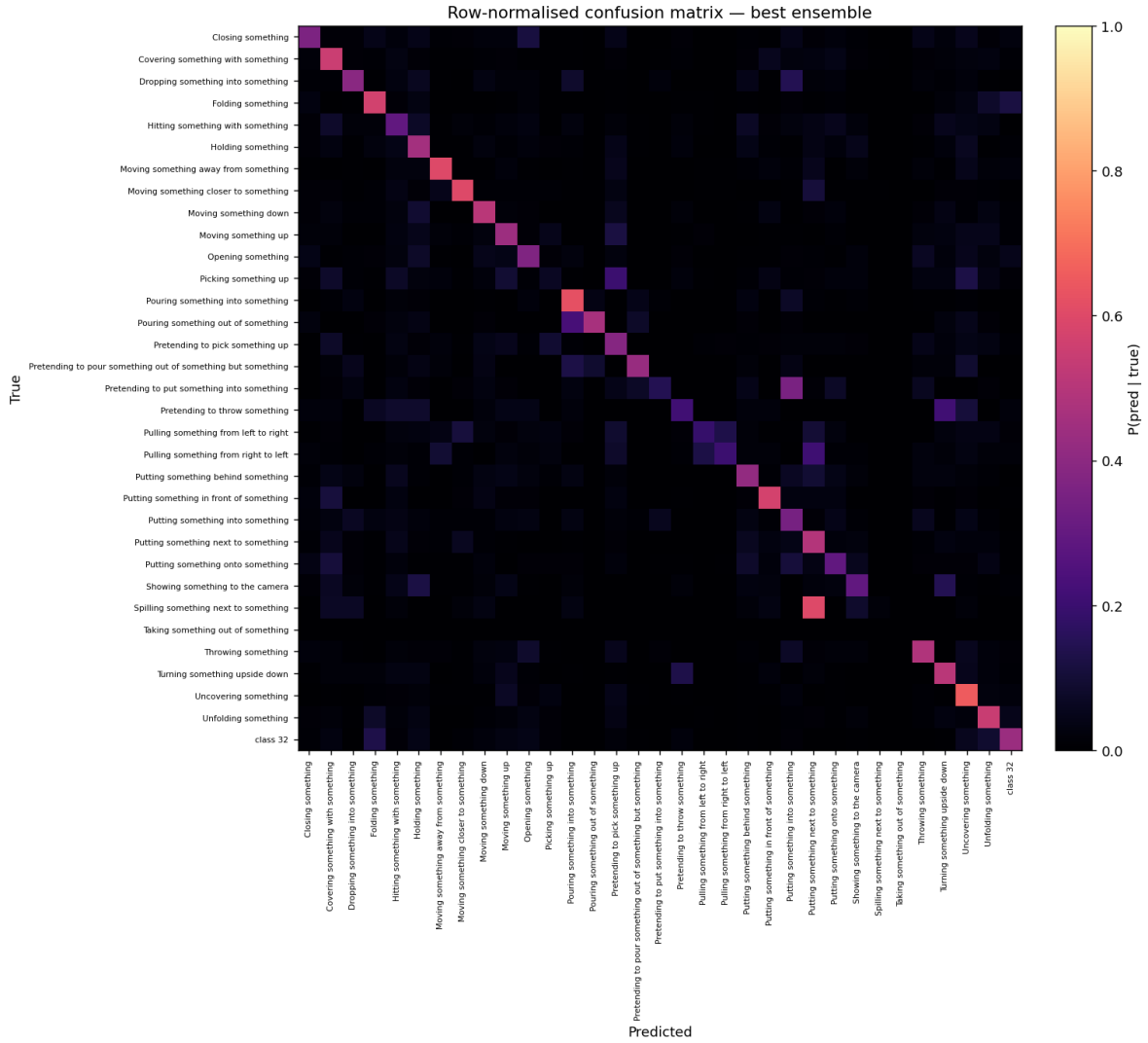


Figure 18: Most-confused class pairs. The dominant errors are intent (real vs. pretended) and temporal-direction confusions (Sec. 3.3).

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